

small blocks and analyse it in parallel using MATLAB.

Peer. Analysis of electrophysiological data is often time consuming. Optimum utilisation of available resources by sharing computational resources among multiple users is the solution for this issue. Analysis could be done dynamically in various MATLAB sessions on a single computer or parallel in multiple computers in the network.

Some noteworthy features

Contrary to the signal-processing tools that have a graphical user interface (GUI), FieldTrip is designed to be used as a toolbox rather than an application. This gives the developer the flexibility not to stick to a specific GUI structure.

Let us take a look at some interesting features of this tool.

No dedicated GUI. Since FieldTrip lacks a dedicated GUI, users need to have direct interaction with MATLAB command-line functions or scripts. Obviously, they need to have some basic knowledge about MATLAB functions, but this allows them to have a combination of various functionalities that are very specific in nature for biomedical analysis.

The toolbox consists of both high-level and low-level functions. The former provides an easy programming structure allowing the user to have a step-by-step detailed analysis. The latter are intended for core functions that are not so user-friendly and cannot be easily used by a common neuroscience researcher. These functions are normally hidden from regular end users in private directories.

Handling data. Yet another important concept in this toolbox is that, data is handled completely in the hands of the users. As data passes through various functions, it gets transformed. So it is the duty of the user to explicitly manage data.

Users should assign meaningful names and save intermediate values as mat files, if they want to use those

Software at a glance

Name of the package: FieldTrip

Type: Biomedical signal-processing toolbox for MATLAB

Main developers: Robert Oostenveld and team at Donders Institute for Brain, Cognition and Behaviour of the Radboud University Nijmegen, The Netherlands

Year of initial development: 2003

Used for: Analysis of non-invasive and invasive electrophysiological data

For more information: www.fieldtrip-toolbox.org

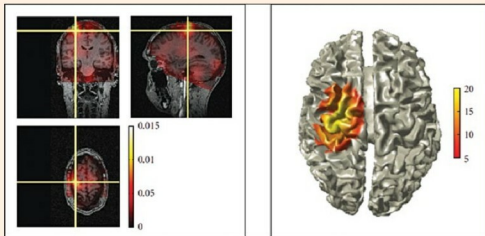


Fig. 3: Visualisation of source reconstructed data

values again. In order to adapt to external software, users can also port the files to non-MATLAB file formats.

Batch processing. With a given set of parameters and a large number of subjects, if users need to analyse cognitive experiments, batch processing is the best choice. It allows them to have flexibility for further re-evaluation with a whole new set of parameter settings.

Result visualisation. MATLAB by itself has a variety of data visualisation and plotting tools. But when we are dealing with complex analysis or visualisation of multidimensional data, this could be difficult. In order to resolve this issue, FieldTrip comes with several high-level plotting functions.

For visualising sensor-level data, it is interpolated on a 2D projection of sensor positions.

Modular design. One major advantage of having a modular design is reusability of the code. fileio module that we discussed earlier is used in both SPM8 as well as FieldTrip software. Not only does this help in avoiding recurrence of work in

rewriting the code, but also provides access to a comprehensive set of data formats. Other modules that are shared between these software are forward and Preproc modules.

While FieldTrip uses SPM for processing anatomical MRI data, SPM relies on FieldTrip for various algorithms on analysis of EEG and MEG data. The codes are also shared with software including EEGLAB, BESA and BCI2000.

Boosting experimental neuroscience

FieldTrip has brought significant contributions in the field of experimental neuroscience. The open source nature of the code has allowed active exchange of ideas between developers and users. The fact that experimental scientists and method developers have benefitted from this tool is exemplified by hundreds of research publications based on the tool. If you are a serious researcher in neuroscience, we suggest you give it a try. ●

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